Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Advanced Polymerized Prodrug Microneedle Patch for Sustained Transdermal Drug Delivery			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (**X**), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (**X**), Quantum ( )			
Industrial Technology Classification [1]	**Technical Field 1** (Large Category)		**Technical Field 2** (Large Category)	
	Bio Medical		Mechanical / Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Biotechnology	Functional Biomaterials	precision manufacturing machinery	optical energy-based processing machines
Concept of Proposed Technology	We have developed an **innovative microneedle patch** (MNP) composed entirely of a **polymerized drug**: specifically, **17β-estradiol** (E2), the major female sex hormone. The macromolecular pro-drug materials are cured thermosets in which every repeating unit in the polyester chains contains an equivalent of E2. These crosslinked polyester resins undergo slow hydrolytic degradation to release **low doses of estrogen, locally at the site of implantation**, for exceptionally **long time periods** (up to many years). Transdermal microneedle patches offer **minimally invasive**, painless administration of estrogen. These delivery methods enhance patient compliance relative to repeated injections with hypodermic needles and they require lower doses than oral administration due by avoiding first-pass hepatic clearance. Of all the many advanced MNP technologies, there are still no examples of MNPs composed entirely of poly(pro-drug). Our lab has developed a new class of polymerizable estrogen pro-drugs containing pendant alkene groups. In this proposal, we will polymerize the pro-E2 monomers into high resolution microneedle arrays. Two key innovations are proposed: First, we will			

	tune the glass transition of the polymer network such that the needles are hard and glassy at room temperature for skin penetration, but then they transition into a soft/rubbery material upon equilibration to physiological temperature. The second is the use of high-resolution direct laser writing methods to create the microneedle arrays.
Rationale for Joint R&D Project	We have developed working prototypes and demonstrated their ability to mechanically puncture skin surrogate materials, as well as quantify the long-term, low-dose hormone delivery. We have an awarded US patent and several high-profile papers published. The work was supported by a NIH grant and led to successful technology generation. Our next step is to identify industry partners to perform pre-clinical de-risking and further commercialize this new biotechnology.
Key Words	1. Polymerized pro-drug biomaterials 2. Vastly extended long-term term drug release 3. Parallelized Direct Laser Microneedle array manufacturing processes 4. Transdermal hormone delivery

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)
R&D Objectives	This project will generate new poly(pro-drug) based microneedle patches capable of releasing low doses of E2 transdermally for exceptionally prolonged timescales. The chemistry and engineering approaches used here will be broadly generalizable in two key ways: 1) our polymerization scheme is easily modified to accommodate nearly any drug molecule with two or more reactive functional group handles, and 2) the MNP of poly(pro-E2) can be applied to a broad range of clinical applications where prolonged low-dose E2 is desired. Our goal is to apply these E2-releasing MNPs in a preclinical rodent model to first assess safety and skin irritation and later to explore the impact of the patch on medical applications where long-term, low dose hormone supplementation is needed.			
R&D Target Specifications	1. **Synthesis and Fabrication**. We successfully developed the polymer chemistry synthetic and fabrication approach to produce arrays of microneedles consisting of densely cross-linked networks of alkene functionalized pro-drugs of 17β-estradiol (here called “estrogen” or “E2”) using two photon lithography. The synthesis of estrogen-based diacrylate monomers proceeded through a one-step esterification of E2 using the anhydride of 2-carboxylethyl acrylate. First, the anhydride was generated via carbodiimide coupling. The crude products could be purified with a simple aqueous workup, making this approach significantly more efficient. Second, 17β-estradiol was reacted with the freshly prepared 2-carboxylethyl acrylate anhydride in the presence of 4-dimethylaminopyridine (DMAP) and triethylamine in anhydrous DCM, to yield a complex mixture of estrogen-based diacrylate monomers. The crude mixture of all monomeric products is a viscous oil that photochemically cures readily, without any purification, but resists premature solidification by monomer crystallization, even when stored at −20°C in the dark (with inhibitor). Microneedle arrays will be fabricated using direct laser writing. A target for the project will be the parallelization of the writing process resulting in resolution specifications of ~1 micron resolutions for critical dimensions, which greatly facilitates penetration of the skin. 2. **Mechanical Properties**. First, we quantify the mechanical properties of monolithic prismatic rectangular bars of the cross-linked poly(pro-drug) material without microneedle molding to assess the inherent, bulk mechanical properties of the materials. Using dynamic mechanical analysis, we can simultaneously obtain the storage and loss moduli across a range of strains, frequencies, and temperatures, which combine to provide a measure of the glass transition (Tg). We find that the plateau storage modulus of the material at room temperature is about 1 GPa, consistent with a material stiffness sufficient to puncture skin. A key goal of this project was to create microneedles for which the Tg lies between room temperature and physiological temperature, which was indeed accomplished. The peak in the tan∂ curve goes through a maximum at Tg ~ 33 ºC. Secondly, we endeavored to characterize the mechanical properties of microneedle arrays composed of poly(pro-E2), in order to confirm the feasibility of skin penetration. These experiments are done in compression with a 1kN load cell where the tips of the needles impinge onto a flat steel plate. We repeatedly compressed the array up to some pre-defined max loadings of increasing force (50-100 N) and discovered that the needles elastically deform and recover reversibly up to about 97 N, at which point failure by needle bend-over is observed. As a result, we estimate the total force per needle is about 9.7N, which is far more than sufficient to penetrate skin according to literature precedence and our own experiments with skin surrogate materials as described below. Penetration of Skin Surrogate Material. Before moving into animals, we tested the ability to manually press the MNP into a stack of parafilm approximately 1 mm thick. We find that the MNP easily is pressed into the			

	parafilm and remains embedded until manually peeled off. Examination of layers shows full penetration of three layers (160 μm each, 480 μm total) and partial penetration/indentation into the fourth layer in the stack, which agrees well with the measured height of ~550 μm height for each individual needle. The holes in the parafilm are visualized by staining with crystal violet and then imaged with a light microscope. The MNP after repeated application (3x insertion, peel-off) on parafilm showed that 99 of the 100 needles remained undamaged. Only one individual needle was found to be missing after the 3rd trial. 3. **Estrogen release** kinetics in vitro. Combination of LC/ms-ms and ELISA assays underway with MNP of polypro-E2 degrading in PBS buffer at 37 ºC, sampled biweekly for up to 6 months. Preliminary calibration curves show: Limit of quantification for the LCMS of ~22 pg is reproducibly achievable. This is more than enough sensitivity: the 10x10 microneedle array contains about 40% estrogen equivalents, by mass, and thus can theoretically produce up to 0.6 μg of pure estrogen upon complete degradation and release.
R&D Plan	The major female sex hormone **17β-estradiol** (E2) is clinically utilized in a broad range of treatments, including menopause, recurring urinary tract infection, birth control, certain cancers, hypogonadism, and transgender therapy. In many cases, pro-drugs or derivatives of E2 are employed to enhance the bioavailability and pharmacokinetics/pharmacodynamics (PK/PD) relative to the native bioactive E2 molecule. The pro-drug approach is supposed to alleviate many problems associated with repeated injections, which suffer from pain and poor patient compliance, or oral administration which requires high doses due in part to first-pass hepatic clearance. **Microneedle patches** (MNP) are a leading emergent technology that enable minimally invasive and painless transdermal drug administration across the stratum corneum. Many groups have explored a variety of MNPs, including those loaded with pro-drug compounds, but to our knowledge MNPs **composed entirely of poly(pro-drug)** have never been fabricated to date. Our groups have pioneered thermoplastic polymerized estrogen as a macromolecular pro-drug scaffold with tunable mechanical properties, hydrophobicity, biocompatibility, and vastly prolonged controlled release of E2 locally at the site of implantation. Here, we propose to develop a new technology for producing thermoset polymerized estrogen in the form of a microneedle patch. **Polymerized drugs**, also known as poly(pro-drugs), are macromolecular biomaterials composed of pro-drug repeating units. These materials undergo degradation to release bioactive compounds over very long timescales (months to years) relative to conventional approaches in which a drug compound is blended or mixed into a biologically inactive polymer carrier or matrix material. Here, the material is both a depot of the active compound, and also serves as its own carrier material. Such poly(pro-drug) biomaterials can elute very low concentrations of drug (nanomolar) as they gradually degrade (most typically via hydrolysis), locally at the site of implantation, for **vastly extended timescales**. Another key advantage of poly(pro-drugs) is their **stellar atom economy**: they inherently possess much higher total drug loading capacity, since each repeating unit within the polymer chain is composed almost entirely of the drug itself on a per atom basis. This poly(pro-drug) approach is in contrast to drug-loading into an inert polymer matrix, in which the drug is only a tiny fraction of the overall scaffold mass. In this initial exploratory grant application, we will focus narrowly on the synthesis and characterization of the MNP with the desired target material properties. If successful, the fundamental engineering advances in this work will lay the foundation for future developments in prolonged E2 delivery for a range of potential uses, to include spinal cord injury (SCI). The results are also highly generalizable to any drug molecule that contains two or more reactive functional groups (e.g. resveratrol, gabapentin, fingolimod, dexamethasone, curcumin and probucol).

	Aim 1. Synthesis and Fabrication. We will prepare a library of poly(pro-estrogen) microneedle patches with systematically tuned chemistries. A pro-drug of estrogen containing photopolymerizable moieties will be dissolved in organic solvent with multifunctional linkers. Direct laser writing, first with two photon lithography and subsequently with highly parallelized direct laser writing using cheaper diode laser sources will be used to form microneedle arrays consisting of crosslinked network material. We will tune the linker length and initial monomer concentration to control network mesh size and thus the resulting thermomechanical material properties. Similar to results we have previously obtained by molding of larger feature size needles, we are specifically targeting networks with a glass transition temperature (T_g) intermediate between room and body temperature (between 25 and 37 ºC) in order to afford materials that are hard and glassy for the initial implantation, but become soft and rubbery once they equilibrate to physiological temperature. The desired T_g is achieved by finely adjusting the chain length of a flexible linker units which connects cross-link junctions in the polymer network. **Aim 2. Characterization.** We will quantify the thermomechanical properties, degradation and release kinetics, and biocompatibility with dermal fibroblasts in vitro. The quality and reproducibility of the MNP by the above fabrication methods will be assessed by scanning electron microscopy (SEM) on at least 9 samples from 3 different batches. Mechanical characterization under compressive loads will be carried out in our proximal probe laboratory, using a modified nanoindentation procedure. We will select needles with a minimum average failure load of 0.5 N per individual needle. We will examine the failure mode (putatively via buckling near the tip) and perform a detailed Weibull analysis of the failure strengths. The MNPs will be characterized by their ability to puncture elastomers that mimic the properties of human skin, under compressive stress, as a function of temperature. The degradation and drug release rates will be quantified in vitro by immersion of the MNP in buffer at 37 ºC for varying times (from 1-6 months) followed by SEM analysis, liquid chromatography/mass spectroscopy (LCMS), and gel permeation chromatography (GPC). **Impact:** We propose the first polymerized estrogen microneedle patch for sustained long-term, low-dose release. A key innovation is tuning the polymer network to exhibit hard/glassy properties during implantation at room temperature, but transitions to soft and rubbery once equilibrated to physiological temperature.

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	Researchers at RPI will lead the chemistry and biological characterization aspects of this work. We will make the pro-drugs and perform the controlled polymerization into microneedles of defined geometry. We will then rigorously characterize the kinetics of degradation and E2 release from these scaffolds. We will quantify their mechanical properties, and skin-penetration ability, as a function of needle geometry, crosslink density, temperature, strain rate, and tip sharpness.
Expected R&R of Korean SME Partner	The partner will perform safety and implantation studies in rodent models of transdermal drug delivery. The partner will quantify hormone blood levels as a function of time post implantation and will monitor animal welfare. Finally, a functional readout for the efficacy of the hormone supplementation in ovariohysterectomized female mice. An ovariohysterectomy (OVH) in female mice is a common surgical procedure for research, removing both ovaries and the uterus to study hormone deficiency, menopause, bone loss (osteoporosis), metabolic changes, or cardiovascular disease, typically done via abdominal incision (ventral or dorsal) with detailed post-op care for recovery, analgesia, and ensuring hormone washout for reliable results, often using specialized protocols for success. With our long-term E2-delivering microneedle patch, we anticipate the ability to address health targets in OVH mice over longer time periods without the need for repeated needle injections.

5. Expected Outcome/Impact

Expected Outcomes and Impact	The major female sex hormone **17β-estradiol** (referred to here as "estrogen" or "E2") is used in a broad range of hormone replacement therapies (HRT), including for menopause, birth control, certain cancers, hypogonadism, and transgender therapy.[1] The global market for HRT was about $14B in 2021 and growing at a compound annual growth rate of 6.1%, according to a report by Fortune Business Insights.[2] One of the biggest drivers of market growth in HRT currently is new advances in **slow-releasing, low-dose delivery methods** that include transdermal gels and **microneedle patches** (MNPs). MNPs were named as one of **Top 10 Emerging Technologies** by the World Economic Forum in 2020.[3] The advantages of MNPs are many: painless local delivery of active compounds, no scarring, and improved patient compliance, relative to repeated hypodermic needle injections.[4] Notably, transdermal delivery requires **lower doses** than oral administration because first-pass hepatic clearance is nil, which further improves safety.[5] Still, some **critical challenges remain** – about 8% of patients with an estrogen-releasing transdermal patch removed it due to skin irritation.[4] Dissolving patches have lower rates of irritation, but the release rate is inherently much faster. Some technologies do include slow-releasing nanoparticles in a dissolving microneedle matrix, but the scalability of such complex manufacturing processes is limited and costly. Emerging areas of potential application for E2 delivery also include novel therapies for spinal cord injury (SCI) and traumatic brain injury (TBI), owning to the neuroprotective and neurotrophic effects of E2 in the injury microenvironment.[6] E2 reduces inflammation, glial cell reactivity, oxidative stress, and glutamate excitotoxic neuronal death, while also providing neurotrophism in the central nervous system (CNS).[7] Since traumatic CNS injuries persist for years,[8] there is an urgent and as-yet unmet need for releasing neuroprotective and neurotrophic drugs for much longer duration.[9] In this proposal, we aim to develop a novel class of MNP materials with precisely controlled long-term E2 release, which have promise for new biomaterials with neuroprotective and neurotrophic effects. Orally administered estrogen suffers from low aqueous solubility, poor bioavailability and high doses are needed due to first-pass hepatic clearance.[10] **Pro-drugs** are drug molecules with a chemical modification to improve some property of pharmaceutical interest (solubility, bioavailability, biodistribution), which are then cleaved in vivo to release the active drug compound.[11] E2 is often administered as a pro-drug, such as hydrophobic ester derivatives in oil or as hydrophilic phosphate and sulfonate conjugates in aqueous conditions. In each case, the pro-drugs are hydrolyzed in vivo to slowly release the active drug. **Pro-drugs of estrogen**, especially esterified estradiols, are widely used clinically as intramuscular injections.[12] These are generally hydrophobic esters that undergo gradual hydrolytic cleavage in vivo to release E2 over time (half-life ~ 1 week). Even still, repeated monthly injections of drug are needed, which is a clear disadvantage. The goal of this proposal is to fabricate polymerized pro-E2 into solid thermosetting microneedles that can effect long-term, low-dose transdermal hormone delivery. Medical applications range from birth control to post-menopausal care, gender-affirming care, spinal cord injury, and veterinary medicine in ovariohysterectomized dogs.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization